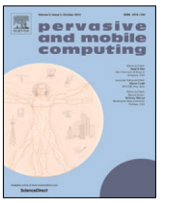
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Detecting structural anomalies of quadcopter UAVs based on LSTM autoencoder



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ABSTRACT

Detecting a structural anomaly, such as a damaged propeller or motor, is crucial for mission-critical operation of unmanned aerial vehicles (UAVs). The existing solutions often fail to detect structural anomalies because the pre-defined parameters required for the solution are limited in reflecting the flight pattern or the external environment, such as wind conditions. In this paper, we propose a method for detecting structural anomalies in quadcopter UAVs, using only regular data and specifically considering flight patterns and runtime flight conditions. To this end, we employ a long short-term memory (LSTM) autoencoder model to learn complex features of regular flight data. While flying the UAV, the trained model estimates the degree of outlierness of the incoming data and assesses abnormal behavior of UAV by adaptively considering its movement. This way, the proposed method accurately detects structural anomalies in UAVs regardless of the runtime environment or flight mission. Our experiment results with an off-the-shelf UAV show that the proposed approach detects diverse structural anomalies by an average of 98.6% specificity and 90.3% sensitivity.

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1. Introduction

Quadcopter unmanned aerial vehicles (UAVs) have recently been used as a popular means of logistics in multiple domains, such as lightweight delivery service, military surveillance, agricultural industries, and so on [1–4]. While conducting such critical missions, UAVs are often exposed in hazardous environments that raise maintenance and safety issues [5,6]. Therefore, monitoring the physical health of UAVs is essential for robust operation of the devices. Improper management of the devices or external physical shocks cause damage in propellers, motors, and frames, resulting in *structural anomalies* which should be dealt with before the UAV fails.

Several attempts have been made to detect physical damages of UAV such as damaged propellers or loosened motor screw in quadcopter UAVs. One approach is to use a physical model that describes the relationship among various UAV subsystems [7–12]. The technique obtains the model parameters in a controlled environment and detects anomalies using the difference between the measured values and the model-estimated values. Another approach is to develop heuristic rules for detecting anomalies [13]. Statistically analyzing the massive amount of flight data, the technique devises a set of rules and applies them to detect anomalies. Both the model-based and rule-based approaches rely on pre-defined parameters obtained manually in advance. Thus, such approaches are not generalized for dealing with UAVs running under varying flight conditions.

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Recently, research has been conducted to exploit machine learning for structural anomaly detection [14,15]. Machine learning-based approaches are divided into three categories according to the algorithm used: reinforcement learning, supervised learning, and unsupervised learning. Reinforcement learning-based anomaly detection enhances its knowledge consistently through appropriate feedback, such as a reward function [16]. As we cannot define all types of structural anomalies in advance, it is hard to find a generally available reward function. Supervised learning-based approaches train models, such as a multi-class neural network [17], random forest [18], and support vector machine (SVM) [19], using normal and anomalous data instances. This method is not feasible practically due to the overhead of labeling every data instance. The method is also limited in detecting unknown types of anomalies. Unsupervised learning methods learn the pattern of regular data solely based on intrinsic properties of the data. Previous works exploited unsupervised learning, such as the isolation forest [20] and the long short-term memory (LSTM) prediction model [21], to detect structural anomalies in UAVs. The works did not, however, consider real flight conditions, resulting in the possibility of erroneous detection.

Even with machine learning-based approaches, many challenges exist and should be solved to develop practically applicable methods for detecting structural anomalies in UAVs. First, we should detect an anomaly using only regular or normal data, fundamentally because of the lack of anomaly data. In practice, a large portion of regular data is obtained from anomaly-free UAVs, leading to an unbalanced dataset between normal and anomaly data. In a controlled environment, certain types of anomaly data can be acquired, but other types of anomalies occur only during real flights. Second, a UAV flight is likely to be affected by external environments such as wind or by the given mission, which would cause noise in the acquired data. For example, the preliminary experiment results in Section 2 show that the flight patterns are affected by various factors, and the runtime flight characteristics of a UAV must be considered to improve the detection accuracy of structural anomalies.

In this paper, we propose a method for detecting structural anomalies of quadcopter UAVs, using only regular data and specifically considering flight patterns and runtime flight conditions. We employ an LSTM autoencoder model to learn complex features of regular flight data. The model is trained to reconstruct the regular flight data with an encoding and decoding process, and used to score the degree of outlierness of the given flight data. The structural anomaly is then detected taking into account the movement of the UAV. Using inertial measurement unit (IMU) sensor data, we calculate the degree of dynamicity of the UAV and adaptively set the decision score, which is the threshold for detecting anomalies. The contribution of our work is as follows:

- We propose an LSTM autoencoder-based method for detecting structural anomalies that is highly practical because it uses only regular data.
- The proposed method considers runtime factors of UAV flights, such as the flight pattern and the environment, leading to better detection of structural anomalies.

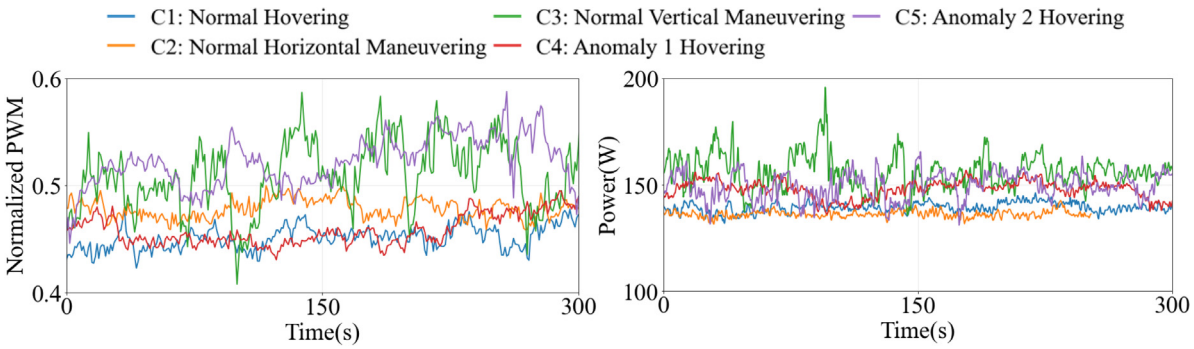
2. Challenges in detecting structural anomaly

We present the challenges in detecting structural anomaly of the UAV and show the limitation of previous work.

2.1. Challenges

We conducted a preliminary experiment, using the DJI Flame Wheel F450 model [22], to investigate the effect of varying environments on flight data and to analyze the anomaly data from broken propellers in particular. Three different types of propellers were used for the experiment: a normal propeller (Normal), a slightly broken propeller (Anomaly 1), and a severely broken propeller (Anomaly 2) (see Fig. 12(a) for images of the propellers). Using normal propellers, we flew the UAV and acquired flight data such as the pulse-width modulation (PWM) values of each rotor, battery-related data, and IMU data with varying movements: hovering, horizontal maneuvering, and vertical maneuvering. The hovering mission has no specific paths except for initial launching and final landing. In the case of horizontal and vertical missions, the UAV repeatedly flies about 10 m vertically and horizontally, respectively. For the broken propellers, we obtained the data only with the hovering mission because of safety issues. We conducted the experiment at a playground in light air wind condition (1–3 knots) to minimize the effects of runtime environments.

Fig. 1 shows the flight data obtained from different flight experiments. Note that although there were vertical movements of the UAV at the beginning and end of the experiment, we plotted the data during hovering for a fair comparison. Fig. 1(a) shows the traces of PWM which indicates the amount of power loaded on each rotor. With vertical maneuvering (C3), the PWM is high and fluctuates severely, compared to hovering (C1) and horizontal maneuvering (C2). Note that gravity could make the power consumption fluctuating on vertical movements even in a normal operation. The trace of Anomaly 1 (C4) shows a similar pattern as the trace of hovering using Normal (C1). The trace of Anomaly 2 (C5) is similar to the trace of vertical maneuvering with Normal (C3) but different from the trace of hovering with Normal (C1). Fig. 1(b) shows the traces of the power consumption for each experiment. C3 shows a large power consumption with a highly fluctuating trace, exhibiting a similar pattern with C5. In the case of C4, despite of its anomaly, its trace is not much different from C1 or C2. Therefore, traces of the anomalous propellers (C4, C5) are barely distinguishable from those of the Normal traces (C1, C2, C3).



(a) PWM trace of different flight data (b) Power trace of different flight data

Fig. 1. Flight data obtained from different flight experiments.

In summary, the flight data are affected by the flight environment (i.e., the given flight mission), and the flight data patterns can differ depending on the movement of the UAV. The results motivated us to consider the flight environment when detecting anomalies. Furthermore, even with the structural anomaly, the acquired data are often difficult to distinguish from the normal data pattern. Because structural anomaly data have complex and sequential patterns, to detect anomalies, we should detect contextual anomalies, not point-type anomalies.

2.2. Limitation of previous work

Anomaly detection has been actively studied in diverse research fields and application domains [14,23]. Although many practical solutions have been proposed such as statistics and machine learning-based methods, domain-specific knowledge is essential for accurate detection of anomaly. Studies on detecting UAV structural anomalies are generally classified into two approaches: the conventional approach and the machine learning-based approach.

Limitation of Conventional Approaches. The conventional methods for detecting structural anomalies are broadly divided into physical model-based approaches and rule-based approaches. Physical model-based approaches use a physical model representing the relationship among subsystems based on physics properties. The approach typically finds various parameters of the physical model through flight experiments in controlled environments. Birnbaum et al. [7] exploited the correlation between UAV control input and UAV movement, and performed statistical analysis through the recursive least squares method. Amoozgar et al. [8] used a two-stage Kalman filter to estimate the possible errors for each actuator. Other approaches constructed a physical model of the sensor by estimating the nonlinear parameters [9,10]. If the value obtained deviates from the model, the method reports an anomaly. The detection accuracy of model-based approaches depends on the accuracy of the model. However, unlike in a controlled environment, a UAV in the wild moves dynamically and is exposed to external environments, such as wind, lessening the model accuracy. Furthermore, the technique requires different model parameters for different types of UAVs, and this leads to practical limitations for generalized use.

The rule-based approach establishes rules for detecting anomalies by statistically analyzing real flight data. Ghalamchi et al. [24] calculated the vibration spectrum of the UAV's rotor and detected large vibrations of the rotor as anomalies. Ghalamchi et al. [25] exploited the correlation between the rotor vibration and the user command to detect anomalies. Since the approach exploit real flight data, this method is possibly more practical for detecting anomalies during real flights. To further understand the method, we implemented one of the rule-based methods [25] and evaluated its detection accuracy. Exploiting two IMUs, a structural anomaly is detected through the correlation between the measured data from each IMU sensor. In the implemented method, an anomaly is reported when the correlation is less than 0.8. Table 1 shows the results. In the case of normal data (C1, C2, C3), the rule-based method works well in Normal hovering (C1) or Anomaly 2 (C5) conditions; however, as the UAV moves dynamically (C2, C3), normal data are reported as anomalous. The results indicate that the rule-based method could fail to detect a less-severe case such as Anomaly 1(C4) in Fig. 1.

Machine learning-based anomaly detection. Many attempts have been made to detect anomalies using machine learning algorithms, because the approach often outperforms conventional methods in many application fields. The approach can also be applied to detect structural anomalies in UAVs. In general, machine learning-based anomaly detection is divided into three categories: reinforcement learning, supervised learning, and unsupervised learning. First, reinforcement learning is suitable for sequential decision-making problems. Lu et al. [16] used a reinforcement learning technique in which the UAV learns to compensate for temperature changes in each rotor and detects sudden temperature changes as anomalies. The method chooses the action at the given state and enhances its knowledge through the observed reward. The detection accuracy relies strongly on the reward; thus, the developer should find the best reward function in a brute-force manner. However, the reinforcement learning method is not suitable for detecting structural anomalies of UAVs as a well-working reward function is hardly obtainable in practice.

Table 1
Correlation of IMU data for various flights.

		Correlation					
Case #	***Flight***	***AccX***	***AccY***	***AccZ***	***GyrX***	***GyrY***	***GyrZ***
C1	Normal hovering	0.93	0.95	0.97	0.98	0.99	0.99
C2	Normal horizontal	0.73	0.92	0.83	0.98	0.98	0.99
C3	Normal vertical	0.81	0.79	0.69	0.98	0.99	0.99
C4	Anomaly 1 hovering	0.93	0.90	0.81	0.98	0.98	0.99
C5	Anomaly 2 hovering	0.11	0.25	0.07	0.93	0.98	0.73

Second, supervised learning learns the relationship between the input and the labeled output; then the trained model is used to detect the anomaly. Misra et al. [18] used random forest, one of the supervised learning methods, which classifies data by constructing multiple decision trees and analyzing the impact of each error on the sensor data. Iannace et al. [17] generated a classification model, by collecting sound from UAV, to detect unbalanced propellers. The support vector machine method was employed to model the motion of the UAV using a gyroscope and an accelerometer as features [19]. This approach assumes that normal and anomalous data are equally available for training. Therefore, the techniques are limited in practice because the training data covering a wide spectrum of structural anomaly are rarely obtainable. The one-class SVM [26,27] could be considered for the solution, which exploits the boundary hyperplane or hypersphere to construct normal boundaries. However, as discussed in Section 2.1, the noise is likely to occur with the dynamic UAV movement, and such noise leads to a sub-optimized model.

Third, unsupervised learning-based methods can be employed to detect anomalies. The approach gathers and learns the anomaly-free data pattern, relying solely on intrinsic properties of the normal data instances without manual labeling. The Such characteristics are well-fitted in our problem where data are collected mostly from anomaly-free UAV operations. Previous works exploit the isolation tree [20] and the LSTM prediction model [21]. The works, however, did not consider runtime factors, such as movements of the UAV or wind conditions, and often failed to detect anomalies with the noisy flight data.

3. LSTM autoencoder-based anomaly detection

3.1. Need for LSTM autoencoder

Many unsupervised learning algorithms are candidates for detecting structural anomalies in UAVs [14]. Deep learning is widely used for detecting anomalies, because the deep neural network learns the pattern with deeply structured layers and generally outperforms the other machine learning techniques. In this work, we specifically employ a deep autoencoder method that belongs to the unsupervised learning category [28]. The method requires only normal data for learning complex features and consists of an encoder and a decoder for reconstructing the given data. Given the input, the encoder compresses it as a low dimensional latent vector upon which the decoder reconstructs the original data. Repeating this process with the regular data, the model learns the compressed representation of the normal data pattern. After being trained, the model should be able to reconstruct the normal data successfully, while failing on anomaly data. The reconstruction error can then be used to determine the degree of outlierness of the given data.

The key to employing a deep autoencoder is selecting the type of deep neural network, for example, a convolutional neural network or a recurrent neural network (RNN) [29]. Because UAV flight data are continuously generated over time, we exploit the RNN-based LSTM model that handles sequential data [30,31]. The unit cell of the LSTM model is composed of the LSTM cell state and three different gates (input gate, forget gate, and output gate) that control the cell state. Each gate uses the current vector and the previous state at every given time step. The input gate memorizes the current vector using the activation function. The forget gate sorts the relevant and irrelevant historical information and feeds only relevant information to the cell state. The output gate calculates the final result using both of the input gate and the forget gate. This process allows the LSTM model to handle sequential data adequately, and thus, is suitable for detecting contextual anomalies [32].

3.2. Learning process overview

Fig. 2 illustrates the overall structure of the LSTM autoencoder used in this work. The LSTM autoencoder learns the complex relationship among k different parameters from sequential flight data. We normalize every k parameter in the dataset to the range of 0 and 1. Then, input sequence s is used as a concatenation of w vectors where w is the number of data instances. The LSTM model uses s_t as an input sequence at given time t, which consists of w data instances from v_{t-w+1} to v_t, and each data instance is made up of k different parameters. Given the input, the encoder of the model sequentially feeds the data instances to each unit cell, obtaining the state of the parameters at every given time

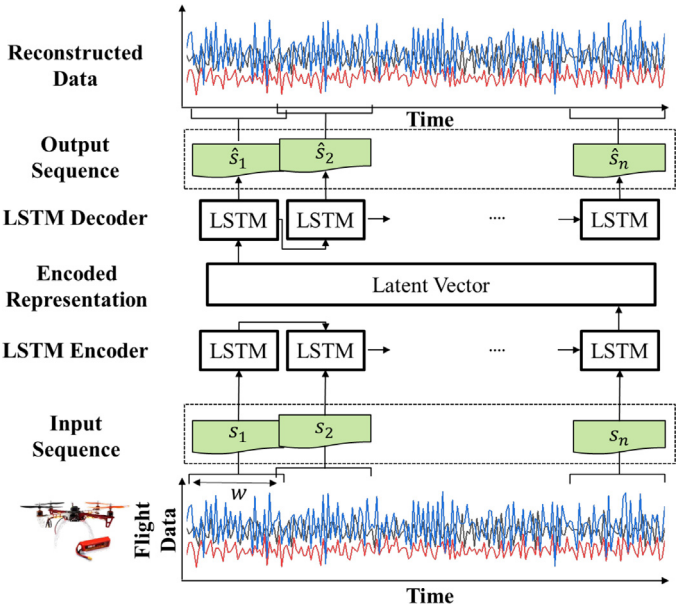


Fig. 2. LSTM autoencoder overview.

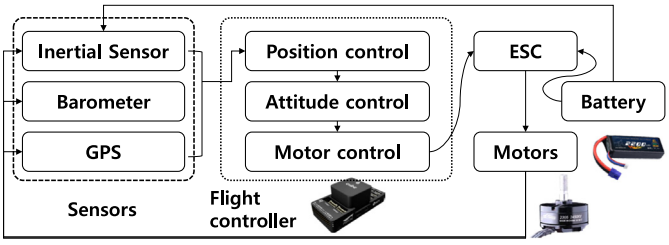


Fig. 3. Simplified UAV operation process.

step. The embedding features from the output gates are fed into the next layer, and this process compresses the input data repeatedly until the latent vectors are obtained which are the lower dimensional representation of the input data. Meanwhile, the decoder proceeds in the opposite direction from the encoder, reconstructing the original sequential data from the latent vector. Let us define $\hat{s}_t$ as a reconstructed output sequence from the decoder. We obtain the reconstruction error which is the difference between original data s_t and reconstructed data $\hat{s}_t$. Using a large amount of normal flight data, the autoencoder adjusts the weight parameters to minimize the reconstruction error, learning complex features within w sequential data.

3.3. Parameter decision

To train the autoencoder model, we should choose the parameters to use as input. A naïve solution is to use all parameters obtained from the flight data. However, the solution could worsen the model's performance due to the curse of dimensionality [14]; therefore, we choose a specific set of parameters considering the operation of the UAV. Fig. 3 illustrates the simplified operation of the UAV. The quadcopter UAV consists of a flight controller, various sensors, the frame, and four pairs of motors and propellers. The UAV gathers flight-related data via sensors, such as the inertial sensor, the barometer, and GPS, and adjusts the PWM for each motor using the calculated results from the flight controller. Then, the electronic speed controllers (ESCs) receive PWM signals to control the motors powered by the on-board battery. This process is repeated until the mission is complete.

We observed that the power drain from the battery is closely related to the overall UAV control process. The output power from the UAV battery is adjusted according to the rotor speed changes. The anomalous rotor would affect the energy efficiency, generating specific patterns in power consumption. Thus, we used the power drain as an input parameter. Meanwhile, the flight controller estimates the position and altitude of the UAV, and then according to the subsequent movement of the UAV, the UAV determines PWM correspondingly. If an anomaly causes the wrong position or altitude, an anomaly in the PWM would occur, and the values are used to detect the structural anomaly. In summary, we exploit

sequential data of power and PWM value of each motor as an input of the model. The model compresses and reconstructs the data while learning the relationship between them.

3.4. Anomaly detection

After training the model, we decided whether the given data are anomalous. Fig. 4 illustrates the overall detection process which is divided into two parts: the scoring algorithm and UAV movement-aware detection. The scoring algorithm is used to assess the degree of outlierness, and the UAV movement-aware detection method adaptively determines the structural anomaly.

Scoring the degree of outlierness. We need to score the degree of outlierness for the decision. The autoencoder model, using the encoder and the decoder, learns the way to reconstruct the given data so that the model outputs the same results as closely as possible. Setting equal weight for the w vectors, we calculated the reconstruction error by using the mean squared errors between the output data and the given data from the time-series data. Because the model is trained with normal data, the input should be reconstructed successfully and should result in little reconstruction error. For the anomaly data, the model should fail to reconstruct the input data and result in a large reconstruction error. Therefore, the reconstruction error is used to score the degree of outlierness.

Algorithm 1: UAV movement-aware anomaly report system

Input: trainData ← regular training data set
S ← real time input sequence with w data instances
$\widehat{G} \leftarrow [g_X, g_Y, g_Z]$ from gyroscope of each axis
$\widehat{A} \leftarrow [a_X, a_Y, a_Z]$ from accelerometer of each axis

```
func reconstructionError(trainData)
    reData ← trainedModel(trainData)
    n ← length(reData)
    diff[1..n] ← 0
    for i ← 1 to n do
        diff[i] ← meanSquaredError(trainData, reData)
    end for
    mean, std ← diff.mean, diff.std
    return mean, std
func movementScoring(Ĝ, Â)
    movement ← Σ_{i=X,Y,Z} (g_i^2(t) + a_i^2(t))
    return movement
/* Main */
mean, std ← reconstructionError(trainData)
while UAV flying do
    Ŝ ← trainedModel(S)
    error ← meanSquaredError(S, Ŝ)
    /* normalization with gravity compensation */
    Ĝ, Â ← normalization([g_X, g_Y, g_Z], [a_X, a_Y, a_Z + 9.8])
    movement ← movementScoring(Ĝ, Â)
    decisionScore ← mean + std × (movement + c)
    if error > decisionScore do
        reportAnomaly(error)
    end if
end while
```

UAV movement-aware detection. As the anomaly data are not easily available, we should set the decision score which is the threshold for the anomaly, using the reconstruction error obtained from the regular data. One possible solution is to use the average and the standard deviation of the errors. The n-sigma rule is one of the solutions for detecting anomalies which reports an anomaly when the given data are not within the n standard deviations of the mean. Although this rule commonly works for various applications, the solution is naïve for use in this problem. As discussed in Section 2, the UAV data pattern varies according to the movement and the flight environment. We should consider the movement of the UAV for the elaborated detection. Fig. 5 illustrates the concept of the UAV movement-aware anomaly detection. The proposed solution adaptively sets the decision score, i.e., the threshold for the anomaly, considering the movement of the UAV. If the reconstruction error is larger than the decision score, an anomaly is reported.

A key idea of the proposed method is to assess the dynamicity of the UAV movement. To this end, we exploit the gyroscope and the accelerometer in IMUs that are readily available in a UAV. The gyroscope sensor measures the orientation and the angular velocity, and the accelerometer measures the rate of change of the UAV velocity. When the device is hovering continuously, no significant changes would be observed in velocity or angular velocity. When the UAV

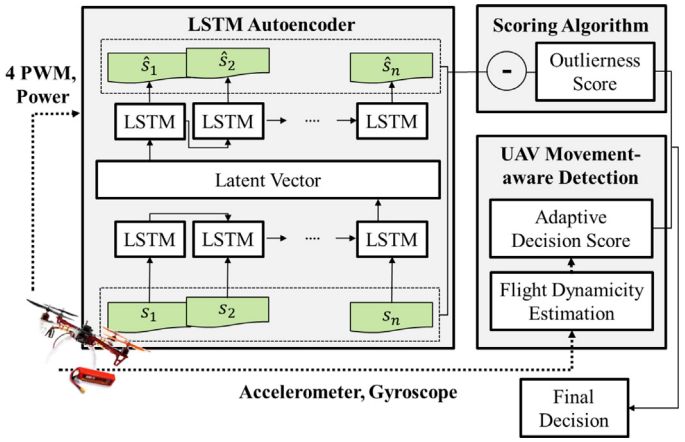


Fig. 4. Overall detection process.

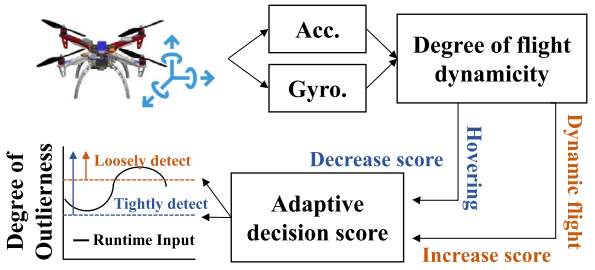


Fig. 5. Concept of UAV movement-aware anomaly detection.

maneuvers dynamically, however, the changes in the measured values are non-trivial. Similarly, the external environment affects the sensor value. If a strong wind shakes the device, the gyroscope or accelerometer values will reflect the sudden movement of the device accordingly.

Algorithm 1 describes the detailed process of the anomaly report system while the UAV is flying. To decide the degree of dynamicity, we used the sensing values of the gyroscope and the accelerometer for the x, y, and z axes. The accelerometer, when the UAV hovers horizontally with the ground, always senses -9.8 m/s^2 for the z-axis due to gravity. We compensate for the gravitational acceleration and normalize the values (line 19). We then assess the degree of dynamicity (lines 10–12, 20) using the following equation:

$$movement = \sum_{i=X,Y,Z} \left(norm(g_i)^2 + norm(a_i)^2 \right), \tag{1}$$

where *movement* is the degree of movement at the given time, *norm* the normalization function, g_i the value from the gyroscope of each axis at the given time, and a_i the compensated value from the accelerometer of each axis at the given time. When we calculated a_i, we assumed that the UAV is hovering or maneuvering vertically, hence the value is compensated for gravitational acceleration on the z-axis. When the UAV is maneuvering horizontally, the UAV is tilted, and the gravitational acceleration affects the acceleration values of all axes. Since we compensate the gravitational acceleration on the z-axis, a large *movement* is estimated while the UAV is maneuvering horizontally. At the given moment, we adaptively set the decision score (line 21) using the following equation:

$$decisionScore = mean + std \times (movement + c), \tag{2}$$

where *decisionScore* is the decision score at the given moment, *mean* the average of the reconstruction error of the training data, *std* the standard deviation of the reconstruction error of the training data, and c a positive constant value. As we set the c as a positive constant, the normal data are reported as normal even when the *movement* is nil. When the UAV flies dynamically and *movement* arises, *decisionScore* will increase, loosely detecting the anomaly. Given the input data, we calculate the reconstruction error (line 17), and if the reconstruction error is larger than *decisionScore* (line 22), the system reports the anomaly (line 23).

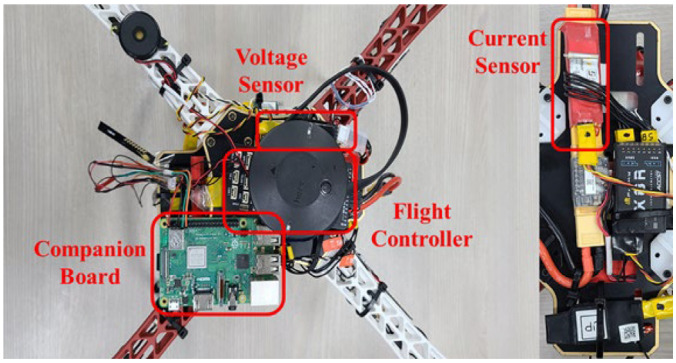
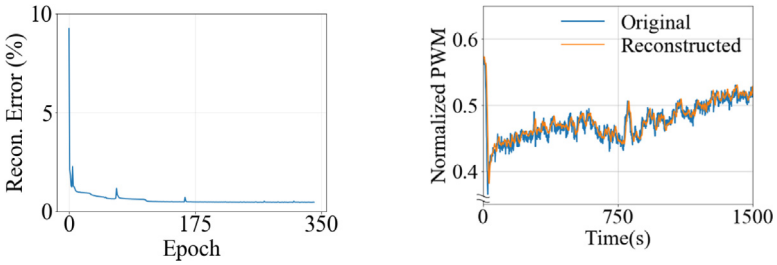


Fig. 6. Assembled DJI F450.



(a) Decreasing training loss (b) Comparing reconstructed PWM

Fig. 7. Model training result.

4. Case study: DJI F450

We conducted a case study to validate the proposed method for detecting structural anomalies. Fig. 6 shows the prototype UAV used for the study. We used the off-the-shelf DJI F450 quadcopter [33] and attached the current and voltage sensors to monitor the power consumption during the flight. The device is powered with a 3300 mAh lithium-ion battery capacity, and the controlling software is Ardupilot open source [34] running on a Pixhawk 2.1 flight controller [35]. The total weight of the constructed UAV is approximately 1.4 kg. For runtime data collection and model inference, Raspberry Pi 3B+ was assembled to the quadcopter. The board was configured for collecting raw data, i.e., voltage, current, PWMs, gyroscope values, and accelerometer values, at 0.1 s interval. The board communicates with the flight controller through the MAVlink protocol. The UAV was operated automatically, without manual control, using the Mission Planner ground control station [36]. For the experiment, the flight missions were preset by mixing the movements of vertical maneuvering, horizontal maneuvering, and hovering. We conducted eight flight scenarios each of which ran approximately 15 min.

The collected data at the quadcopter were transferred to the remote server for model training. Concerning the training of the LSTM autoencoder model, we used sequential data of four PWMs and power consumption as input, each of which was 10 consecutive data instances collected for 1 s. We then normalized the input vector using min–max normalization [37], scaling the values to the range of 0 and 1. The encoder was constructed with four LSTM layers. The first layer had 64 LSTM cells, the second 32 cells, the third 16 cells, and the fourth 8 cells. The decoder was also constructed with four layers. The first layer had 8 cells, the second 16 cells, the third 32 cells, and the fourth 64 cells. We used the mean squared error as a loss function and implemented the model using the TensorFlow framework. Then, we trained the model using the training data. Fig. 7(a) shows the decreasing and converging reconstruction error for the model over several training epochs. The model learns complex features of original regular flight data as the epoch increases. Concerning the training overhead of the model, for good training, about 230 epochs of training were needed for the loss to converge. Training one epoch takes less than one minute on a server with two GPUs, thus it takes less than 4 hours for training the autoencoder model. The small reconstruction error means that the encoder layer extracts the data feature appropriately, and by using it, the decoder successfully reconstructs the original data as shown in Fig. 7(b).

We set the decision score using only the reconstruction error of the normal data set. The average error was 0.00192, and the standard deviation was 0.00233. Because the data set consists of regular data only, the larger c in (2) leads to high accuracy. However, if c were set large, the decision score would increase, and subsequently, the method could frequently fail to detect the anomaly. In this regard, we experimentally set c to 0.82, aiming to judge 99% of the normal data correctly.

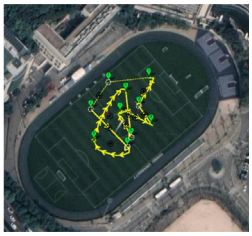
The trained model is converted to the TensorFlow Lite framework and ported on the companion board (Raspberry pi 3B+). The board calculates the reconstruction error at runtime with data collected during the flight. The reconstruction



(a) Horizontally maneuvering

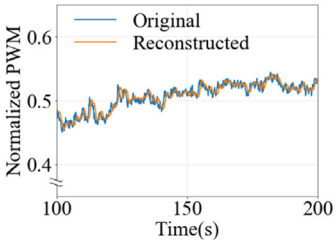


(b)Vertically maneuvering

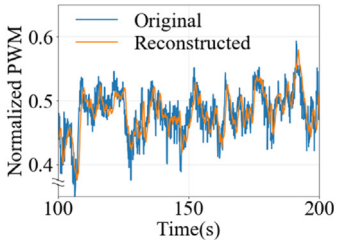


(c) Unrestricted maneuvering

Fig. 8. Flight mission setup.



(a) Reconstruction with normal hovering input



(b) Reconstruction with normal vertical movement input

Fig. 9. Reconstruction with normal input sequence.

error is then compared to the decision score to finally determine if the UAV has an anomaly. When an anomaly is detected continuously, the UAV is forced to land automatically. All the acquired data and results are stored in the companion board for further evaluation.

5. Evaluation

We evaluated the performance and efficacy of the proposed method, using the UAV and LSTM autoencoder model described in the previous section.

5.1. Setup for normal operation

We gathered the flight data, using the anomaly-free UAV, with several types of missions in various flight environments to validate whether the method correctly assesses normal data. To cover the various flight conditions, we conducted flight experiments in three different wind conditions according to the Beaufort scale [38]: light air, moderate breeze, and fresh breeze. Light air means a wind of 1–3 knots, moderate breeze 9–14 knots, and fresh breeze 16–20 knots. We obtained the wind condition data from the public weather service. For each wind condition, four different missions were assigned; hovering at the same altitude, maneuvering horizontally at the same altitude, maneuvering vertically while changing altitude, and unrestricted maneuvering. As shown in Fig. 8, horizontal mission is to maneuver in a rectangle shape at the same altitude. Vertical mission maneuvers at the same position while changing the altitude. Unrestricted maneuvering flies with changes in both position and altitude. We conducted the experiment five times at an outdoor playground for each condition and acquired a total of 80 sets of flight data.

5.2. Decision score

We evaluated how accurately the proposed solution reported regular data as normal. To this end, we first checked if the autoencoder model reconstructed the original data correctly. Fig. 9 shows the normalized PWM traces of the original data and the reconstructed data. In the case of normal hovering, Fig. 9(a) shows that the reconstruction error is small due to the close correlation between the traces. When the UAV moves vertically (Fig. 9(b)), the difference becomes large compared to the hovering case. We evaluated the effectiveness of the adaptively set decision score in coping with varying reconstruction errors. Fig. 10 shows the distribution of the reconstruction errors of the autoencoder model and the average decision score for various conditions. We observed that the reconstruction error increased based on the wind condition or the movement of the UAV. Specifically, the average of the reconstruction error with the fresh breeze was approximately three times larger than the case of light air. When the UAV hovered or maneuvered horizontally, the reconstruction error was relatively small and resulted in a small standard deviation. In the case of the vertical movement,

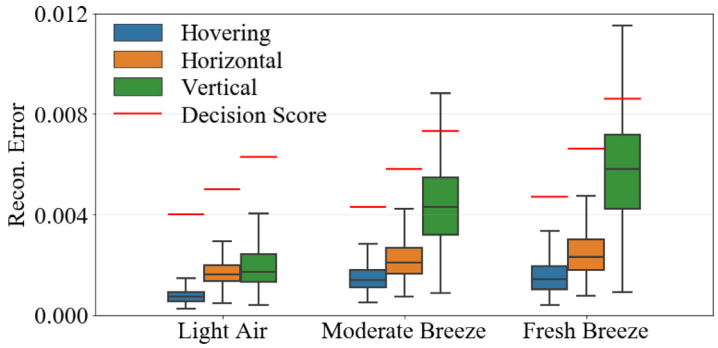
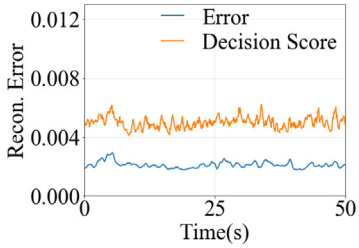
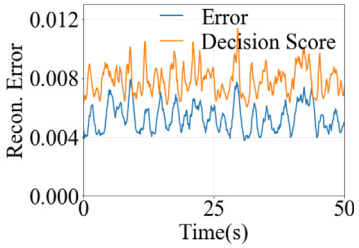


Fig. 10. Distribution of the reconstruction errors and the average decision scores with various conditions.



(a) Decision score for hovering data with light air



(b) Decision score for vertical movement data with fresh breeze

Fig. 11. Errors and decision scores with two types of wind.

Table 2
Decision accuracy of various conditions.

Flight pattern	Wind (knot)			
	Light air (1–3)	Moderate breeze (9–14)	Fresh breeze (16–20)	Total (%)
Hovering	99.8	99.9	99.9	99.9
Horizontal	100.0	99.6	98.6	99.4
Vertical	99.1	96.3	94.1	96.5
Unrestricted mission	99.6	99.5	96.6	98.6
Total (%)	99.625	98.825	97.3	98.6

the reconstruction error was high with a large standard deviation, meaning that noise data occurred frequently. To summarize, the reconstruction error varied depending on the runtime condition. Exploiting the UAV movement-aware detection method, we could correctly judge the normal data despite the increased error.

Table 2 shows the decision accuracy in the various conditions. The average accuracy rate for all data was 98.6%, when c in Eq. (2) was set to target 99% accuracy for the normal data. In the case of light air where less noise occurs, the accuracy rate was higher than 99%. When the UAV flew with a fresh breeze, the wind rocked the UAV, resulting in 97.5% accuracy. We also investigated the effect of the flight pattern on accuracy. The flight data from the hovering mission were stable with little fluctuation, and the proposed method showed high accuracy. In the case of vertical movement, the PWMs and power consumption fluctuated and resulted in 96.5% accuracy. Lastly, the accuracy of the unrestricted maneuvering mission was 98.6% which is higher than that of a vertical mission and lower than that of a horizontal mission. Overall, the accuracy generally decreased in a harsh environment, i.e., fresh breeze or vertical movement, but at least 94% accuracy was acquired in all cases. In short, exploiting the UAV movement-aware detection method, we could correctly judge the normal data regardless of the runtime UAV environment or flight mission.

We further investigated how the decision score we set adaptively copes with harsh conditions. Fig. 11 shows the reconstruction error and the decision score in different flight patterns and windy environments. In the case of hovering with light air, the reconstruction error is small with little fluctuation as shown in Fig. 11(a). The UAV is hardly rocking in this case, leading to a stable decision score. Fig. 11(b) shows the case of vertical movement with fresh breeze. Because the UAV maneuvers dynamically and rock considerably due to the wind, many noise data exist, and this increases the reconstruction error. The reconstruction error at around 30 s is almost two times larger than the average reconstruction error. We found that the decision score changed adaptively according to the movement of the UAV and correctly reported most of the normal points. While existing methods decide the anomaly statically, the proposed method detects an anomaly



Fig. 12. Structural anomalies setup.

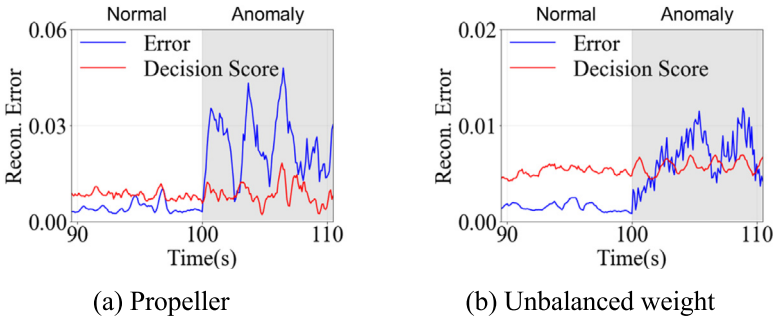


Fig. 13. Anomaly detection scenarios.

by setting the decision score loosely in a harsh environment and tightly where there is little movement. This validates that the proposed method deals adequately with the runtime environment and the flight pattern of the UAV.

5.3. Anomaly setup

Obtaining anomaly data during a flight is practically hard due to many safety issues. To validate the proposed method, we intentionally set, to the extent that failure did not occur, different types of structural anomalies. Fig. 12 shows the structural anomalies we created on purpose: three propeller anomalies, a motor anomaly, and a frame anomaly. Fig. 12(a) shows the three types of propeller anomalies. Anomaly 1 and Anomaly 2 are the same propellers used in the preliminary experiment, and we constructed Anomaly 3 in which almost 60% of the blade is cut off. For the motor anomaly, we loosened three of four screws in one specific motor as shown in Fig. 12(b). To emulate the unbalanced weight state, we attached two types of objects on a leg as shown in Fig. 12(c). Using these six types of anomalies, we set up two experiments using the hovering and unrestricted maneuvering missions (including horizontal, vertical, and diagonal movements). We then conduct flight experiments two times each and acquired 24 anomaly data with a total 360 minutes of flights.

5.4. Anomaly detection

We evaluated two specific scenarios to see how the proposed method detects an anomaly in runtime. Since the intentional damaging of the UAV in the air is dangerous, we emulated abnormal situations by using the mixture of the normal data obtained in Section 5.1 and the anomalous data we acquired from real flight experiments (Section 5.3). We then evaluated the proposed method by using the combined data. Fig. 13 shows the reconstruction error and the decision score for the two scenarios.[1] The non-shaded area in the graph is for normal data, whereas the shaded area is for anomaly data. The first scenario is assumed to occur when the propellers are broken due to a bird strike, for instance, which commonly occurs in real flights [39]. As shown in Fig. 13(a), the reconstruction error in the non-shaded area is lower than the decision score. The reconstruction error increases around at 97 s, but the decision score also increases, and the system correctly assesses the normal data. When a bird strike is assumed to occur at 100 s, the UAV rocks momentarily; thus, the reconstruction error and the decision score increase. The error is larger than the decision score; therefore, the method successfully reports a structural anomaly.

In the second scenario, we assumed that a weight imbalance occurs on the device while the UAV is performing a logistics mission. This scenario is likely to occur when the parcel box is not properly attached to the UAV. Fig. 13(b) shows

[1] We evaluated two scenarios of damaged propellers and unbalanced weight because the anomaly pattern of a loosened motor is similar to the case of damaged propellers, causing overload on a specific rotor.

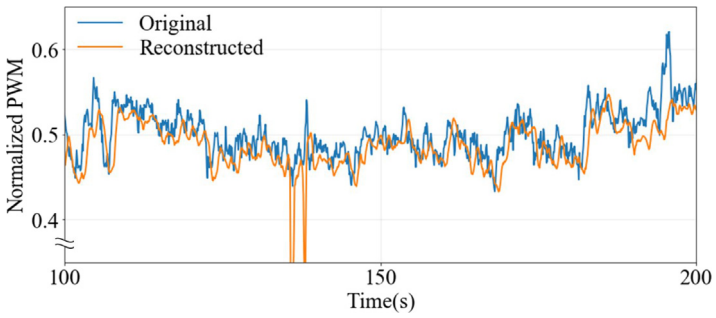


Fig. 14. Reconstruction with input sequences of structural anomaly.

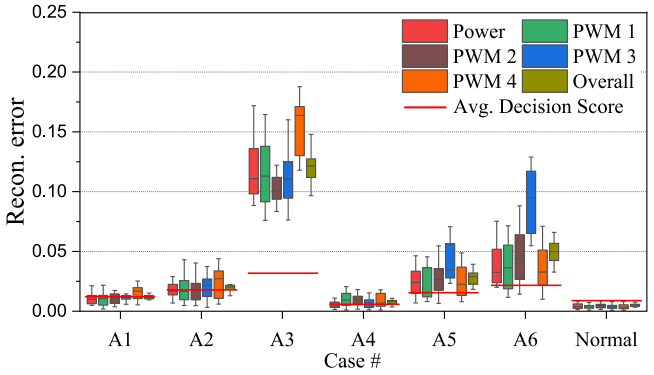


Fig. 15. Distribution of reconstruction error while hovering.

Table 3
Accuracy of various anomalies while hovering.

Case #	Anomaly	Accuracy
A1	Propeller Anomaly 1	0.762
A2	Propeller Anomaly 2	0.950
A3	Propeller Anomaly 3	1.000
A4	Loosened motor screw	0.653
A5	Unbalanced weight (100 g)	0.999
A6	Unbalanced weight (200 g)	1.000
Overall		0.894

the detection scenario in which anomaly data are obtained while attaching a 100 g object on a leg. Similar to the first scenario, the reconstruction error is lower than the decision score in the non-shaded part. However, in the shaded area, the PWM of the overloaded rotor increases, which leads to an overall increase in the reconstruction error. The proposed method quickly and accurately detects the anomaly.

We further evaluated how accurately the proposed solution detects various anomalies. We first validated whether the reconstruction error for anomalous flight data was larger than that for the regular flight. Fig. 14 shows the normalized PWM traces of the original data, and the reconstructed data obtained from the anomalous UAV. The difference between the original data and the reconstructed data is clear, in comparison with the reconstruction of normal data in Fig. 9. The overall results from the hovering experiment are summarized in Table 3, and the detailed distribution of the reconstruction error and the mean of the decision score at each case is shown in Fig. 15. The result from the normal hovering experiment is presented as Normal on the x-axis.

The damaged propeller makes small lift and unbalanced revolutions per minute (RPM) in each rotor, making the UAV rock. The UAV keeps controlling the PWM to cope with the rocking, and this leads to the anomaly pattern. When the UAV flew with Anomaly 1, the anomaly was detected with an accuracy of 76.2%. A more severe anomaly such as Anomaly 2 or Anomaly 3 caused the UAV to rock severely and resulted in the large reconstruction error for A2 and A3 shown in Fig. 15. Specifically, the propeller error for Anomaly 3 is too large to be compared to other cases. Because of the large reconstruction error, the method could easily detect the A3 with 100% accuracy. The motor anomaly from the loosened screws made the UAV shake, but the degree was small compared to other cases. As the flight was stable because we set the

Table 4
Accuracy of various anomalies while maneuvering.

Case #	Anomaly	Accuracy
A1	Propeller Anomaly 1	0.789
A2	Propeller Anomaly 2	0.994
A3	Propeller Anomaly 3	1.000
A4	Loosened motor screw	0.683
A5	Unbalanced weight (100 g)	1.000
A6	Unbalanced weight (200 g)	1.000
Overall		0.911

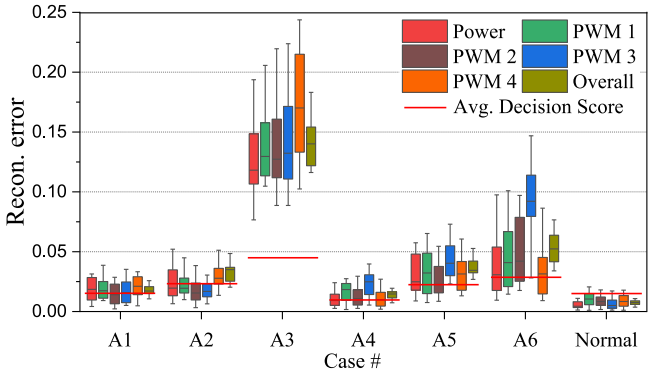


Fig. 16. Distribution of reconstruction error while maneuvering.

degree of anomaly weakly in this case, the reconstruction error was smaller than in the other cases, as shown in Fig. 15. The method correctly reported the motor anomaly with the lowest accuracy, 65.3%, among the six anomalies. However, considering that the method generates only a 1.5% false alarm rate when normal data are given, the detection rate of 65% is meaningful for reporting structural anomalies. The unbalanced weight anomaly made the specific rotor overloaded. A large reconstruction error occurred with the PWM of the overloaded rotors (PWM 4), while the PWM of the other rotors showed a small error. The reconstruction error increased with the object weight, and the method successfully detected these anomalies with little failure.

Table 4 summarizes the detection accuracy of various anomalies while the UAV is maneuvering, and Fig. 16 shows the distribution of reconstruction error and mean decision score in each case. The overall accuracy is improved compared to the detection accuracy from the hovering experiment in Table 3. As shown in the figure, the mean decision score in each case has increased, over the score in the same case in Fig. 15, due to the dynamic movements of UAV. This means that the proposed method loosely detected the anomaly while the UAV was maneuvering. However, the increase in the reconstruction error is larger than the increase in the decision score. The structural anomaly such as broken propellers causes less propelling power than the case of anomaly-free rotor, and the PWM value of the anomaly rotor rapidly increases while the UAV moves upward. Specifically, the UAV was severely rocking during vertical movements in the case of A3, and the device turned over eventually. The LSTM autoencoder model failed to reconstruct the original data with the anomalous flight patterns, and subsequently resulted in significantly larger reconstruction error than the decision score.

5.5. Dynamic runtime environment vs. Structural anomaly

The previous results show that both the dynamic runtime environment and the structural anomaly lead to high reconstruction error. To clarify how the proposed method detects the anomaly, we compared the flight data from the dynamic environment with the data from anomaly 4 which showed the lowest accuracy. Fig. 17 shows the average reconstruction error and the decision score. Figs. 17(a) and 17(b) show the traces when the UAV is hovering and maneuvering, respectively, with fresh breeze. Fig. 17(c) shows the traces of the propeller anomaly 4 (A4). In the case of hovering with fresh breeze, the reconstruction error often spikes due to the intensity and direction of the wind. Consequently, the decision score alters temporarily despite little movements of UAV. The reconstruction error and the decision score of the maneuvering mission (Fig. 17(b)) fluctuate significantly, in comparison with the hovering case. The results show that the proposed UAV movement-aware anomaly detection successfully judges the normal data in the dynamic runtime environment. Fig. 17(c) shows that, although the UAV was hovering, the loosened screws made the UAV shake. Thus, the average level of decision score is high compared to Fig. 17(a). However, the reconstruction error is constantly larger than the case in Fig. 17(b).

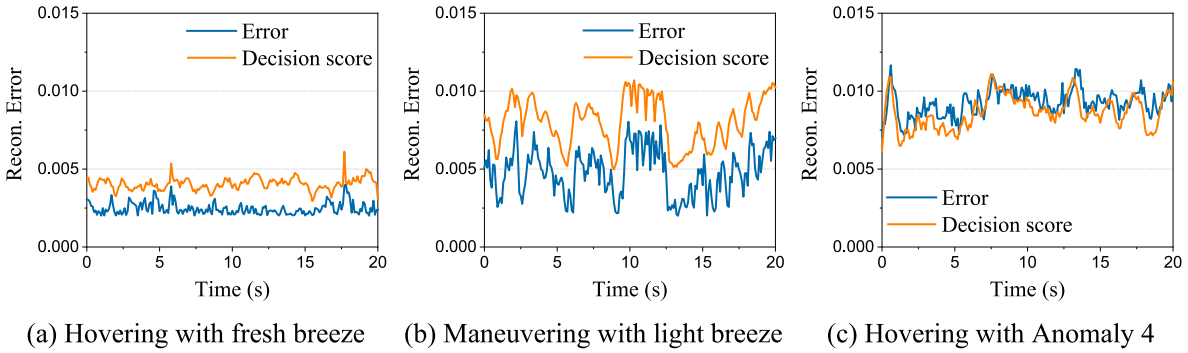


Fig. 17. Reconstruction error and decision score of three different cases.

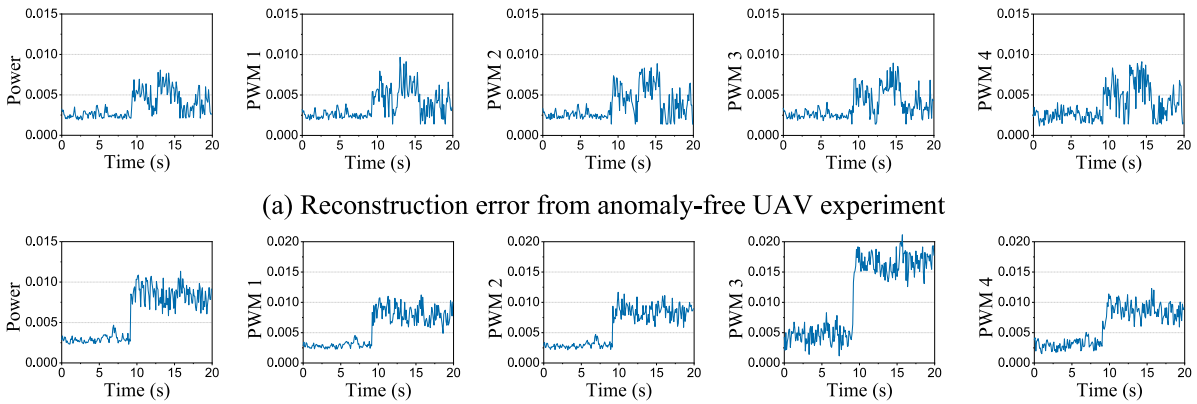


Fig. 18. Distribution of the reconstruction error at each case while maneuvering.

To further analyze the root cause of high reconstruction error in Fig. 17(c), we compared the reconstruction error of the output power and PWMs. Fig. 18(a) shows the flight traces of the anomaly-free UAV. The traces from 0 to 9 s were gathered while the UAV was hovering, and from then, the UAV dynamically moved upward and downward. Fig. 18(a) shows that the reconstruction error of each parameter is evenly distributed regardless of the movements of UAV. Fig. 18(b) shows the traces of flight data of the UAV with anomaly 4. The UAV was configured to conduct the same mission as the normal UAV. The UAV rocked during the mission due to the loosened screw, and the average level of reconstruction error from all the parameters are slightly higher than the error in Fig. 18(a). Specifically, the reconstruction error of PWM 3 continuously shows large reconstruction error compared to the other parameters. The reconstruction error spikes with the movement of UAV. This spiking feature in reconstruction error allows the proposed method to detect the structural anomaly better than the case with the hovering mission.

5.6. Parameter analysis

Input Parameter. We conducted experiments to show that the parameters used in the proposed method, i.e., PWMs and power, are appropriate choices. To this end, we implemented n-sigma rule-based methods using three different parameters: all the sensor values available in the flight controller (i.e., Quaternion, magnetic, velocity, position, power, PWMs), the gyroscope and the accelerometer, and PWMs and power consumption. We set n as a value where the method correctly judged 99% of the normal training data. Table 5 shows the results. *Specificity* represents the rate of correctly identified normal data, and *Sensitivity* the ratio of correctly detected anomalies [40]. *Balanced Accuracy* is the average of specificity and sensitivity. After obtaining the specific n of each method, we evaluated the detection accuracy using the data we acquired in the experiments described in Sections 5.1 and 5.3. The specificity of the training data set was close to 99% in all cases. The result is intuitive because we set n to include 99% of regular data in the training data set. In reality, as an anomaly is likely to lead to the overall failure of the UAV, achieving high sensitivity is more important than specificity.

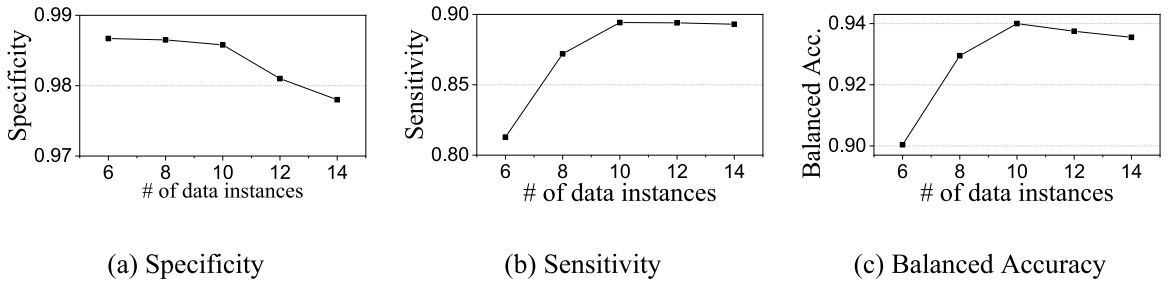


Fig. 19. Experimental results with the different number of data instances.

Table 5
Statistics with other parameters.

Method	Specificity	Sensitivity	Balanced accuracy
Gyro & Accel	0.9910	0.3045	0.6477
All sensor values	0.9906	0.7815	0.8710
PWMs and power	0.9883	0.7958	0.8920
Proposed method	0.9858	0.8942	0.9400

The gyroscope and accelerometer-based method had the lowest sensitivity. The raw sensing value, due to the dynamical UAV movement, had noise, resulting in the model failing to learn the pattern of the regular data. Although more parameters were given as the input, the all sensor value-based method showed low balanced accuracy compared to the PWM and power approach. This is because the high dimension of the data hindered learning of the pattern of the regular data. The sensing data from the gyroscope and the accelerometer are meaningful parameters representing the movement of the UAV. We further compared the *n*-sigma rule method and the proposed method to show the need of considering the UAV movement. Table 5 summarizes that the UAV movement-aware detection improved the detection performance. The naïve *n*-sigma solution set the decision score based on all data without considering the change in the pattern caused by the UAV movement. Such a loose decision score could not detect a subtle structural anomaly such as Anomaly 1.

Impact of Input Data. We use four PWMs and power consumption as input, each of which was 10 consecutive data instances. The amount of sequential data could affect the overall accuracy. We investigated the accuracy of the proposed approach using the data we acquired from the experiments described in Sections 5.1 and 5.3. Fig. 19 shows the results with different size of data instances. Fig. 19(a) shows that specificity decreases with increasing size of data instances. As the size of instances increases, the amount of information for reconstruction increases, which leads to lowered specificity. Fig. 19(b) shows that sensitivity increases as the size of data instances increases, and then converges after 10. Since the proposed system detects anomaly based on the context of sequential data, short input sequence could not contain enough information for detecting structural anomaly. Lastly, Fig. 19(c) shows that the case with 10 data instances achieves the best performance in terms of balanced accuracy.

5.7. Comparison with other methods

We compared the detection accuracy of the proposed method with other unsupervised learning-based methods. Specifically, we implemented the isolation forest-based method [20], the LSTM prediction model-based method [21], and MAD-GAN [34]. The characteristics of anomaly data are inherently different from those of normal data. The isolation forest exploits this characteristic and isolates all the data using a binary tree. The LSTM prediction model uses the LSTM model similar to the proposed method, but there is a difference in the learning method. Our LSTM autoencoder model learns the representation of the given data by encoding and decoding the original data, whereas the LSTM prediction model learns to predict the next value. The model detects an anomaly if the residual between the predicted value and the measured value exceeds the normal range. MAD-GAN uses a generative adversarial training to solve the multivariate anomaly detection problem. The model has a generator that produces fake data similar to the input, and a discriminator that determines whether data is real or fake. The anomaly decision score is determined by adding the restoration loss obtained through the generator and the discrimination loss obtained through the discriminator. We evaluated the proposed method and the relevant methods by comparing specificity, sensitivity, and inference overhead. We loaded each model on the companion board, Raspberry pi 3B+, then simulated the flight experiments by exploiting the real flight data described in Sections 5.1 and 5.3.

Table 6 summarizes the sensitivity, specificity, balanced accuracy of the methods. The table also shows the average execution time for a single inference of each model on the Raspberry Pi 3B+ companion board. The runtime capability of model inferencing is important as the companion board is usually constrained in its computing capability. The isolation

Table 6
Statistics with other methods during flight.

Method	Specificity	Sensitivity	Balanced accuracy	Inference time (s)
Isolation forest	0.7315	0.6139	0.6727	0.01
LSTM prediction model	0.9783	0.4614	0.7198	0.02
MAD-GAN	0.5872	0.9636	0.7754	0.67
Proposed method	0.9841	0.9052	0.9447	0.06

forest showed a low performance in terms of balanced accuracy. The method simply detects outliers rather than learning the data pattern, failing to report normal data correctly in harsh conditions or to detect subtle anomalies. The LSTM prediction model showed high specificity with low sensitivity. Because the prediction model is trained to predict the value of the next step, the method detects only point-type anomalies. As shown in Section 2, a structural anomaly leads to a contextual anomaly in flight data; thus, the LSTM prediction model barely detects such anomalies. In the case of MAD-GAN, the specificity was low compared to other methods, and the average inference time was 0.67 s which is significantly longer than the sensing interval of 0.1 s. With the method, the sensing data will be frequently missed for the inference, resulting in misjudging of normal cases as anomaly.

In summary, the proposed method outperforms the existing unsupervised learning-based anomaly detection techniques for UAV. The method correctly reports almost 99% of normal data as normal, and even with a minor structural anomaly, 65% of the anomalies were detected. Furthermore, the proposed method is light enough to conduct model inferencing in real-time.

6. Limitation

To validate the proposed method, we evaluated several types of frequently occurring structural anomalies based on our experience of flying UAVs. The method was proven to detect predefined types of structural anomalies successfully but does not cover the various types of structural UAV anomalies, presented by Kandaswamy et al. [13]. In this aspect, the present work is limited to detecting only a subset of general anomalies observed in UAV flights. Next, the accuracy of the proposed solution may be degraded if the training data set includes anomaly data. Gathering normal flight data has the potential of obtaining anomaly data from the UAV with undetected structural anomalies. These anomalous data affect the decision score, and the method might fail to detect a subtle anomaly. Last, we chose the DJI F450 UAV with which we can customize both the software and hardware. Certainly, the real off-the-shelf UAVs are hard to customize, hence the proposed scheme is unlikely to be implemented directly. For example, the DJI Phantom 4 or Mini 3 UAVs provide sufficient hardware resources to implement the proposed method, but the devices are proprietary and the flight data is encrypted. Generally speaking, if the commercial UAVs provided software development kit for offering flight data, the anomaly detection models would be running at the ground station. Alternatively, the device manufacturers themselves could adopt the proposed method by running the autoencoder model on their devices.

7. Conclusion

We proposed a practical method for detecting structural anomalies in UAVs solely from regular flight data. Exploiting the LSTM autoencoder, we adaptively decide the structural anomaly of a UAV by considering its runtime movement. Because the decision score is determined by considering UAV movements, an anomaly is detected regardless of the runtime environment or flight mission. The evaluation results showed that the proposed method detects several types of anomalies with high accuracy.

As future work, we plan to apply this technique to many other cases of UAV deployment. This will allow us to figure out the types of frequently occurring structural anomalies, and based on this work, we will be able to extend the current work of detecting anomalies, to its prognosis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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